* Lagrange Multiplier:

To motivate the method of Lagrange multipliers, we are trying to maximize the function $f(x,y)$ subject to the constraint $g(x,y)=0$. Geometrically, this means that we are looking for a point (x_0, y_0) on the graph of constraint curve at which $f(x,y)$ is as large as possible. Then we set:

$$\vec{\nabla} f(x_0, y_0) = \lambda \vec{\nabla} g(x_0, y_0).$$

for some scalar λ. The same condition holds at points on the constraint curve where $f(x,y)$ has a minimum. The scalar λ is called a lagrange multiplier. And the method is called the method of Lagrange multiplier.

Example-1: Use Lagrange multiplier to find the maximum and minimum value of f subject to the given constraint. Also find the points at which these extreme value occurs: $f(x,y) = xy$; $4x^2 + 8y^2 = 16$.

CS CamScanner

Solution: Given that: $f(x,y) = xy$.

subject to the constraint $g(x,y) = 4x^2 + 8y^2 - 16$.

Now, Gradient of f, $\vec{\nabla} f = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j}\right)(xy)$

$$= y\hat{i} + x\hat{j}$$

Gradient of g, $\vec{\nabla} g = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j}\right)(4x^2 + 8y^2 - 16)$

$$= 8x\hat{i} + 16y\hat{j}.$$

Applying the method of Lagrange multipliers:

$$\vec{\nabla} f = \lambda \vec{\nabla} g$$

$$\Rightarrow (y\hat{i} + x\hat{j}) = \lambda(8x\hat{i} + 16y\hat{j}) \longrightarrow (1)$$

Equating the coefficients we get:

$$y = 8\lambda x$$

$$x = 16\lambda y.$$

There are three unknowns but two equations. We can add the constraint to find out the unique solution.

$$y = 8\lambda x$$

$$x = 16\lambda y$$

$$4x^2 + 8y^2 - 16 = 0.$$

From the first two equations:

$y = 8\lambda x$	$x = 16\lambda y$
$\Rightarrow \frac{y}{8x} = \lambda$	$\Rightarrow \lambda = \frac{x}{16y}$

CamScanner

$\therefore \frac{y}{8x} = \frac{x}{16y} \Rightarrow y^2 = \frac{1}{2}x^2 \Rightarrow x^2 = 2y^2$

From the constraint:

$4(2y^2) + 8y^2 - 16 = 0$

$\Rightarrow 16y^2 = 16$

$\therefore y = \pm 1$

When $y=1$, $x^2 = 2 \Rightarrow x = \pm\sqrt{2}$

When $y=-1$, $x^2 = 2 \Rightarrow x = \pm\sqrt{2}$

$\therefore$ The points are $(\pm\sqrt{2}, 1)$, $(\pm\sqrt{2}, -1)$

$f(\sqrt{2}, 1) = \sqrt{2}$

$f(-\sqrt{2}, 1) = -\sqrt{2}$

$f(\sqrt{2}, -1) = -\sqrt{2}$

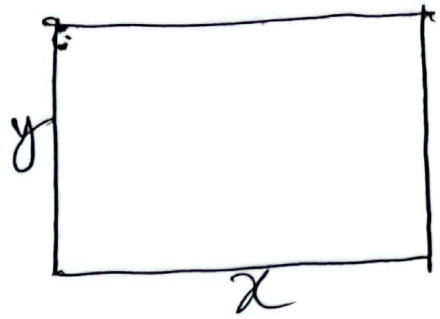
$f(-\sqrt{2}, -1) = \sqrt{2}$

The maximum value of the function is $\sqrt{2}$ at $(\sqrt{2}, 1)$ and $(-\sqrt{2}, -1)$ and the minimum value is $-\sqrt{2}$ at $(-\sqrt{2}, 1)$ and $(\sqrt{2}, -1)$.

<u>Ex-2</u>: Use the method of Lagrange multiplier to find the dimension of a rectangle with perimeter p and maximum area.

CamScanner

Solution: Let:



x = length of rectangle.

y = width of rectangle.

A = area of rectangle.

We want to maximize $A = xy$ on the line segment $2x+2y=P$, $0 \leq x, y$

Here, $f(x,y) = xy$

$g(x,y) = 2x+2y$

Now, $\vec{\nabla} f = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j}\right)(xy)$

$= y\hat{i} + x\hat{j}$

$\vec{\nabla} g = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j}\right)(2x+2y)$

Setting: $\vec{\nabla} f = \lambda \vec{\nabla} g$

$\Rightarrow (y\hat{i} + x\hat{j}) = (2\lambda\hat{i} + 2\lambda\hat{j})$

We get by calculating the coefficients of the like powers: $y = 2\lambda$ and $x = 2\lambda$.

These two equations contain two variables. To get the unique solution we need to add another equation. So, we add the constraint to solve for unique solution.

CamScanner

$$y = 2\lambda$$
$$x = 2\lambda$$
$$2x + 2y = P$$

From the first two equations: $x = y$.

Then: $2x + 2x = P$

$$\Rightarrow 4x = P$$
$$\therefore x = \frac{P}{4}$$
$$\therefore y = \frac{P}{4}$$

So, the dimensions are $x = \frac{P}{4}$ and $y = \frac{P}{4}$.

<u>Circle</u>: A circle is a set of points in the plane whose distance from a fixed point in the plane is constant. The fixed point is the center of the circle and the constant distance is the radius.

The standard form equation of circle is

$$(x-h)^2 + (y-k)^2 = a^2$$

which is centered at (h,k) with radius a.

Example: The equation of circle is given by $x^2 + y^2 - 12x - 16y + 19 = 0$. Find the center and radius of the circle.

CamScanner

Solution: Given that:

$$x^2+y^2-12x-16y+19=0$$

Let's transform the equation into the standard form:

$$\because x^2+y^2-12x-16y+19=0$$

$$\Rightarrow x^2-2.x.6+6^2+y^2-2.y.8+8^2+19=6^2+8^2$$

$$\Rightarrow (x-6)^2+(y-8)^2=100-19$$

$$\Rightarrow (x-6)^2+(y-8)^2=81$$

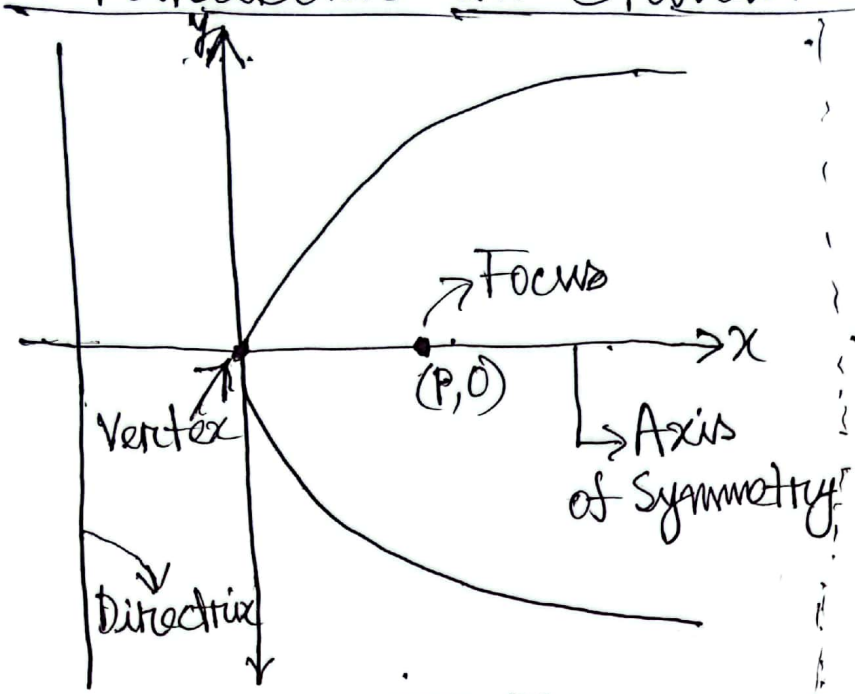
$$\Rightarrow (x-6)^2+(y-8)^2=9^2$$

$\therefore$ The center is $(6,8)$ and radius is 9.

*<u>Parabolas</u>: A parabola is a set of points that consists of all the points in a plane equidistant from a given fixed point and a given fixed line in the plane. The fixed point is the focus and the fixed line is the directrix. A parabola is symmetric about the line that passes through the focus at the right angles to the directrix. This line is called the axis of symmetry of the parabola. The point at which

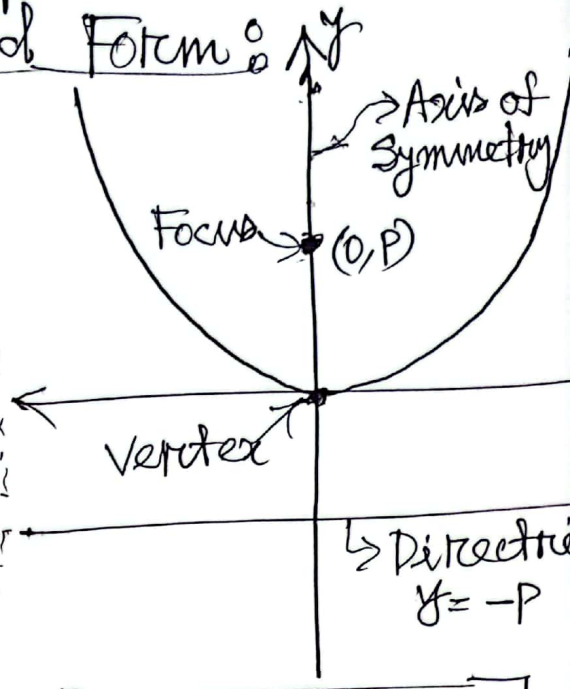
the line intersects the parabola is verte

* Parabolas in Standard Form:



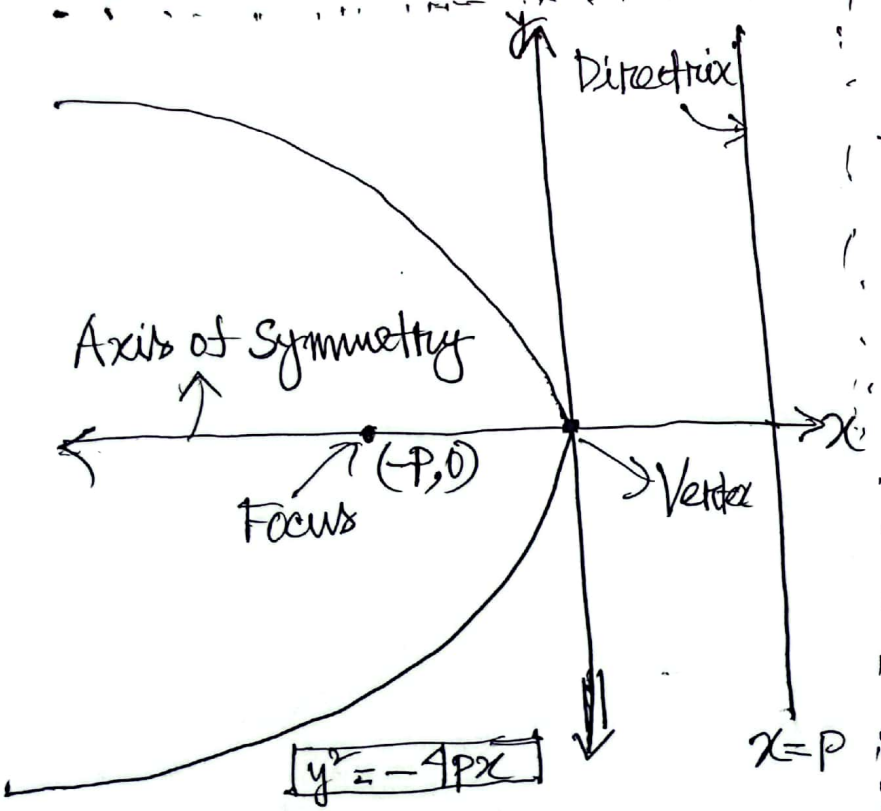
$\boxed{y^2 = 4px}$

Vertex (0,0)
Focus (P, 0)
Directrix: $x = -P$



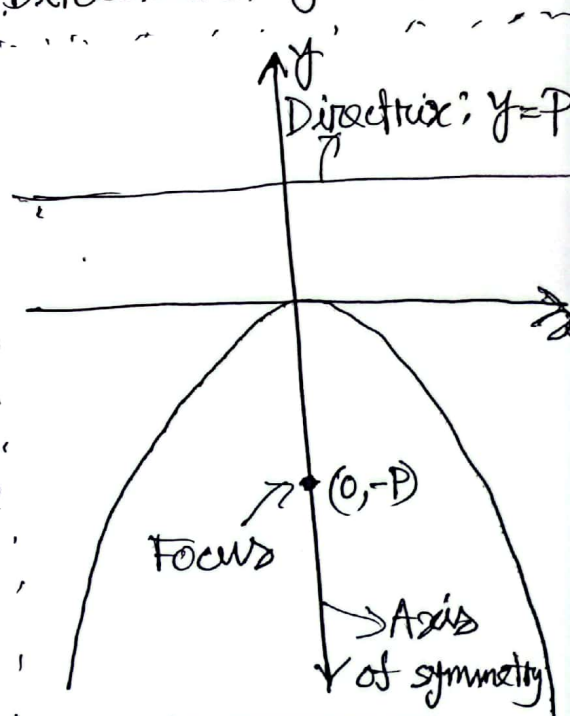
$\boxed{x^2 = 4py}$

Vertex (0,0)
Focus (0, P)
Directrix: $y = -P$



$\boxed{y^2 = -4px}$

Vertex(0,0)
Focus(-P,0)
Directrix: $x = P$



$\boxed{x^2 = -4py}$

Vertex (0,0)
Focus (0,-P)
Directrix: $y = P$

CamScanner

* Standard Equation of Parabola:

The general equation of parabola is given by

$$(y-k)^2 = 4p(x-h) \quad \text{——} \quad (1)$$

or $$(x-h)^2 = 4p(y-k) \quad \text{——} \quad (2)$$

Here, (h,k) denotes the vertex:

Then Focus $(h+P, k)$
Directrix: $x = h-P$ } —— For (1)

and Focus $(h, k+P)$
Directrix: $y = k-P$ } —— For (2)

Example: Find the Vertex, Focus and Directrix of the parabola: $2y^2+3y-4x-3=0$

Solution: Given that:

$$2y^2+3y-4x-3=0$$

$$\Rightarrow y^2+\frac{3}{2}y-2x-\frac{3}{2}=0$$

$$\Rightarrow y^2+2.y.\frac{3}{4}+\left(\frac{3}{4}\right)^2 = 2x+\frac{3}{2}+\left(\frac{3}{4}\right)^2$$

$$\Rightarrow \left(y+\frac{3}{4}\right)^2 = 2x+\frac{3}{2}+\frac{9}{16}$$

$$\Rightarrow \left(y+\frac{3}{4}\right)^2 = 2x+\frac{33}{16}$$

$$\Rightarrow \left(y+\frac{3}{4}\right)^2 = 2\left(x+\frac{33}{32}\right)$$

which implies the standard form of parabola.

CamScanner

Here, the vertex is $\left(-\frac{33}{32}, -\frac{3}{4}\right)$.

From the standard form: $4p = 2$

$$\Rightarrow p = \frac{1}{2}$$

Focus: $(h+p, k) = \left(-\frac{33}{32} + \frac{1}{2}, -\frac{3}{4}\right) = \left(-\frac{17}{32}, -\frac{3}{4}\right)$

Directrix: $x - h = -p$

$$\Rightarrow x = h - p$$

$$\Rightarrow x = -\frac{33}{32} - \frac{1}{2}$$

$$\therefore x = \frac{-49}{32}.$$

CS CamScanner